Contemporary Issues In Big Data

Company - Chugai Pharmaceutical co., ltd.

Submitted by:

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ABSTRACT

This research conducts a thorough investigation into the utilization of big data analytics within the pharmaceutical sector, emphasizing key aspects such as processes, benefits, obstacles, and ethical implications. The study explores various analytical approaches, like descriptive, diagnostic, predictive, prescriptive, and cognitive analytics, and assesses their impact on operational outcomes and decision-making efficiency within the industry. Furthermore, it highlights big data analytics' potential to enhance pharmaceutical research and development by outlining three major empirically-supported and industry-backed benefits. Lastly, it offers strategies for addressing challenges associated with big data analytics to aid pharmaceutical companies in their successful adoption.

The research further explores ethical, legal, and social considerations related to big data analytics within the pharmaceutical industry, tackling matters such as data privacy, regulatory adherence, and data sharing protocols. Through an examination of multiple ethical dilemmas, the study delivers valuable analysis for stakeholders navigating the ethical intricacies of big data analytics. By providing empirical data and industry benchmarks, it illustrates the potential of big data analytics to advance pharmaceutical research and development, resulting in enhanced patient outcomes and state-of-the-art healthcare practices. This report acts as a thorough resource for academics, practitioners, and policymakers seeking to understand the role of big data analytics in the pharmaceutical realm.

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1. Introduction:

Chugai Pharmaceuticals is leveraging digital technologies to drive innovation in drug discovery and improve efficiency across its operations. The company is committed to advancing digital solutions and using big data to create a comprehensive digital platform that includes both software and hardware. Additionally, Chugai is working to establish a robust global IT infrastructure by combining internal data resources and partnering with the Roche Group to develop an analysis platform (Chugai Pharmaceutical co., ltd., 2023).

The practice of pharmacovigilance, which monitors and evaluates drugs to identify and prevent adverse effects or problems throughout their life cycle, is having trouble adjusting its frameworks and regulations to keep up with the expanding amount of data and new methods of big data analysis. Current pharmacovigilance initiatives are highly reliant on healthcare professionals and the pharmaceutical sector, making it challenging yet essential to gather patient-specific details about disease burden and psychology from healthcare databases to better comprehend diseases (Matsuda et al., 2021).

Patients provide valuable information on drug usage, adverse events, and the suffering these incidents cause. This data offers nuanced insights that are difficult to obtain from existing healthcare databases and can help Chugai Pharmaceuticals better understand disease epidemiology, treatment progression, and overall burden. The rich data includes big data sources suitable for assessing patient circumstances (Matsuda et al., 2021).

In addition to other data analysis software, Chugai Pharmaceuticals utilizes SAS software (version 9.4). MeCab, a Japanese open-source program, is used to process unstructured text, which involves segmenting Japanese text without spaces between words, known as morphological analysis. The company also employs R statistical software (version 3.6.2) for text mining and data visualization-related tasks (Matsuda et al., 2021).

2. Application of Big Data Analytics in Chugai Pharmaceutical Co., Ltd.:

2.1. AI-Leveraging Drug Discovery:

Chugai Pharmaceuticals has introduced "DataRobot", a predictive and automated machine-learning platform, into multiple sectors like sales, manufacturing, and research. One of its

applications involves determining the most suitable manufacturing conditions for plants. By utilizing machine learning, the platform can identify these optimal parameters, which is a challenging task when using conventional techniques. This integration of machine learning helps minimize the chances of developing a formulation that falls short of specified criteria (Chugai Pharmaceutical co., ltd., 2023).

Chugai Pharmaceuticals utilizes FRONTEO's AI-powered natural language analyzer, "Amanogawa" (trademark pending), combined with "Concept Encoder (conceptencoder®)" and "Cascade Eye," a distinctive technique that represents disease mechanisms as graphical pathways. This method's objective is to accelerate the drug development process, raise success rates, and make the entire procedure more streamlined (Chugai Pharmaceutical co., ltd., 2020).

Chugai Pharmaceuticals uses artificial intelligence (AI) to reshape multiple aspects of drug discovery. A notable example is their support of the AI Machine Learning x Antibody (MALEXA®) initiative, which applies AI to select therapeutic antibodies for drug discovery projects. By evaluating extensive patterns in antibody amino acid sequences, Chugai rapidly identifies potential candidate sequences. Chugai has further established the effectiveness of machine learning in antibody optimization, specifically for sequence generation and predictive modelling. As a result, the approach can suggest sets of antibody sequences based on preferred medicinal attributes like binding activity, pH dependence, and physical traits (Chugai Pharmaceutical co., ltd., 2023).

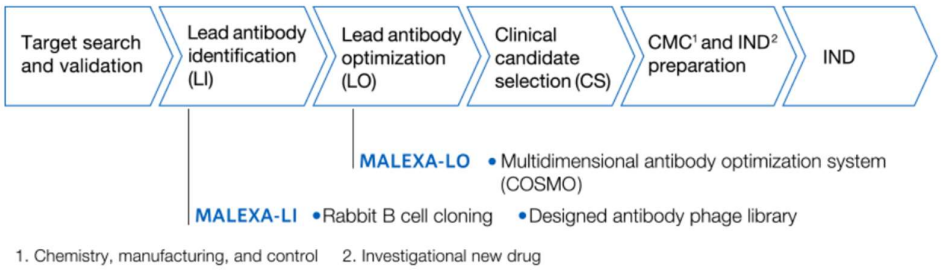

Figure 1: Innovating Drug Discovery Processes Using AI

2.2. Antibody Engineering Technologies:

Chugai has not only created the first-ever therapeutic antibody in Japan but has also built a complete drug discovery platform that includes its exclusive techniques for antibody engineering. Among these are the ART-Ig® technology for producing bispecific antibodies,

the SMART-Ig® Recycling Antibody technology, and the advanced Switch-IgTM Switch antibodyTM technology that resolves safety challenges associated with difficult targets (Chugai Pharmaceutical co., ltd., 2023).

A study prepared by Chugai Pharmaceuticals explains the engineering steps used to develop SKY59, including converting a rabbit antibody into a humanized version and boosting its ability to bind C5. SKY59 was optimized for maximum clinical potential by combining custom variable and constant regions. Later parts of the report delve further into this engineering process (Sampei et al., 2018).

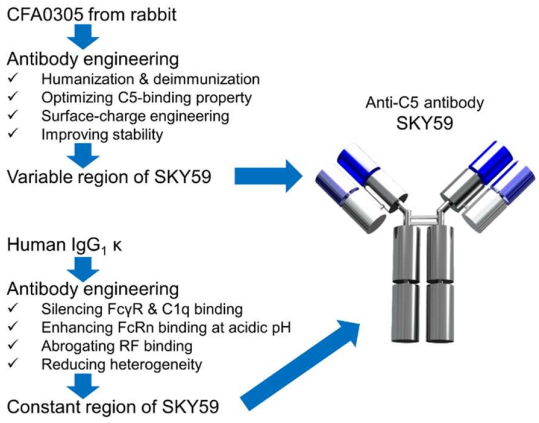

Figure 2: Antibody engineering to generate SKY59

2.3. Automation in Drug Discovery Research:

Chugai Pharmaceuticals and Omron Corporation have collaborated to introduce automation in research processes, with the aim of enhancing drug discovery research efficiency, improving experimental data quality, and supporting new experimental undertakings. This partnership involves setting up a system at Chugai Life Science Park Yokohama for verification tests and basic drug discovery research activities. The system integrates artificial intelligence (AI) and robotic technologies to automate several tasks. Mobile robots will handle the transportation of laboratory samples and management of equipment, while dual arm robots will operate various experimental tools. Automation enables the system to function even during unattended periods, like overnight, ensuring uninterrupted drug discovery experiments. This alliance signifies a notable development in the integration of AI-supported drug discovery and robotic technologies into research methodologies, offering substantial improvements in research

efficiency and quality of results ("Chugai Pharmaceuticals and Omron Collaborate to Enhance Automation in Drug Discovery Research," 2023).

3. Key benefits derived from using big data analysis in Chugai Pharmaceutical Co., Ltd.

3.1. Real-World Data (RWD):

Real-World Data (RWD) is anticipated to elevate patients' quality of life by offering a deeper comprehension of clinical care and outcomes, as well as contributing to the sustainability of social security systems. RWD's versatility allows for its utilization in tasks such as insurance reimbursement cost-effectiveness analyses, prescription cost negotiations, and expediting big data application approvals for clinical drug development. In Japan, there is a persistent dialogue regarding the necessity to cultivate an environment that facilitates the aggregation and assessment of high-quality RWD that can be effectively employed in clinical research (Chugai Pharmaceutical co., ltd., 2023).

Assessing medical data, such as Real-World Data (RWD), can lead to various valuable findings (Chugai Pharmaceutical co., ltd., 2023):

- A better understanding of diseases in basic medical studies.
- Streamlined drug approval procedures and clinical development processes.
- Early recognition of health risks related to drugs and medical devices.
- Evaluating and responding to national health policies.
- Enhanced patient access to medical care.
- Improved quality and effectiveness of nursing care.

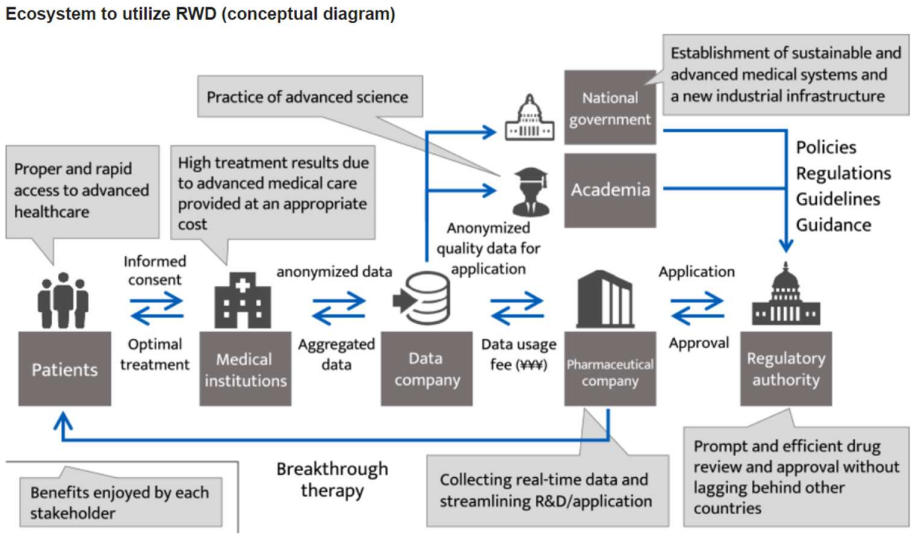

Figure 3: Utilization of real-world data (RWD)

3.2. Digital Biomarkers:

In various clinical development initiatives, Chugai employs wearable technology to collect and analyse ongoing, objective physiological data from patients. Our goal is to examine previously inaccessible real-time crucial data in order to create early proof of concept in drug development. This includes focusing on patient selection, safety management, and digital endpoints. Moreover, our development of dBM is designed to enhance Quality of Life (QOL) following treatment, prevent disease, and offer new approaches for early diagnosis and better understanding of diseases (Chugai Pharmaceutical co., ltd., 2023).

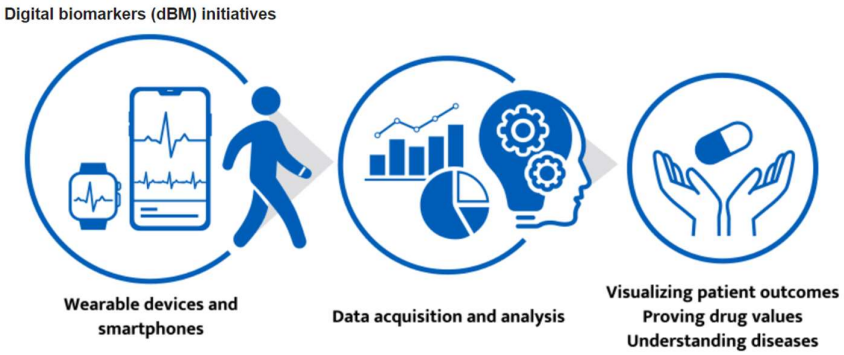

Figure 4: Chugai's dBM initiatives

The assessment of digital biomarkers considers the possible use of big data in clinical trials. Digital biomarkers can serve as indicators of target engagement, with any changes possibly reflecting the effect of bromocriptine therapy. Furthermore, tau and amyloid β-protein in cerebrospinal fluid samples will be studied in conjunction with minimally invasive biomarkers obtained from digital technologies, positron emission tomography, and plasma samples to gain significant knowledge about the disease (Repurposing bromocriptine for Aβ metabolism in Alzheimer's disease (REBRAnD) study, 2021).

3.3. Efficient Data Flow and Collaboration:

Recognition is growing for the value of integrating data from finished clinical trials and sharing big data to advance clinical cancer research and development. Large-scale data sharing from clinical trials has become vital in this effort. ARCAD-Asia, a key initiative that commenced in 2021, actively gathers data from clinical studies performed throughout Asia. Individual patient data (IPD) from ARCAD-Asia is regularly transferred and incorporated into the ARCAD database at the Mayo Clinic. Chugai Pharmaceuticals holds a central position in the ARCAD project (Takeda et al., 2023).

Creating a data sharing strategy can build trust and confidence in clinical trial results by improving the openness of study procedures and reporting. By offering an ideal platform for data sharing, innovative ideas may be spurred, resulting in a more streamlined system with minimized redundancy. Even with the development of various systems to address numerous requests, there remains a gap between the desire for big data and the actual execution of data sharing (Takeda et al., 2023).

4. Challenges faced by Chugai Pharmaceutical Co., Ltd.:

4.1. AI-Based Drug Discovery Efficiency:

The pharmaceutical sector is prioritizing cost-effectiveness and faster new drug development in response to escalating costs. Given the limited availability of target genes for drug development and the increasing intricacy of discovery processes, optimizing resources and increasing success rates are critical priorities. The industry is executing strategies to expedite

processes, lower costs, and streamline workflows to attain these goals, ensuring patients receive advanced treatments promptly and affordably (Chugai Pharmaceutical co., ltd., 2023).

AI, automation, and real-world data innovations are transforming the landscape of drug research and development. AI's influence on disease target identification and therapeutic molecule design is substantial, expediting the process and increasing efficiency. AI-powered analysis of real-world data and digital biomarkers is driving improvements in clinical development decisions. AI further simplifies Proof of Concept determinations by examining early clinical data, streamlining the entire process and accelerating the rollout of novel treatments to patients (Chugai Pharmaceutical co., ltd., 2023).

4.2. Strengthen the Digital Platforms in Clinical Trials:

Clinical trial operations involve extensive documentation and processes for pharmacological approval applications, execution, planning, and implementation. NTT DATA is tackling this challenge by creating an automated document generation system based on protocols, with the goal of improving process efficiency. NTT DATA is partnering with Chugai to conduct concept verification, evaluating the system's efficacy and pinpointing areas for improvement. Additionally, NTT DATA is developing a full-scale clinical trial platform connecting clinical sites and pharmaceutical companies. This platform harnesses cutting-edge IT technology to accelerate new drug discovery by streamlining various clinical trial procedures (Chugai and NTT DATA Complete Demonstration of AI-based Clinical Trial Efficiency Solution : Media | NTT DATA, 2023)

The solution provides a systematic and thorough approach to generating various clinical trial-related documents as part of a comprehensive clinical trial platform. It creates documents such as patient informed consent forms, statistical analysis plans, case report forms for patient trial data entry, and clinical study reports summarizing trial outcomes. These documents are generated using AI technology in conjunction with ontology and semantic components, utilizing data sourced from protocols as input information (Chugai and NTT DATA Complete Demonstration of AI-based Clinical Trial Efficiency Solution : Media | NTT DATA, 2023).

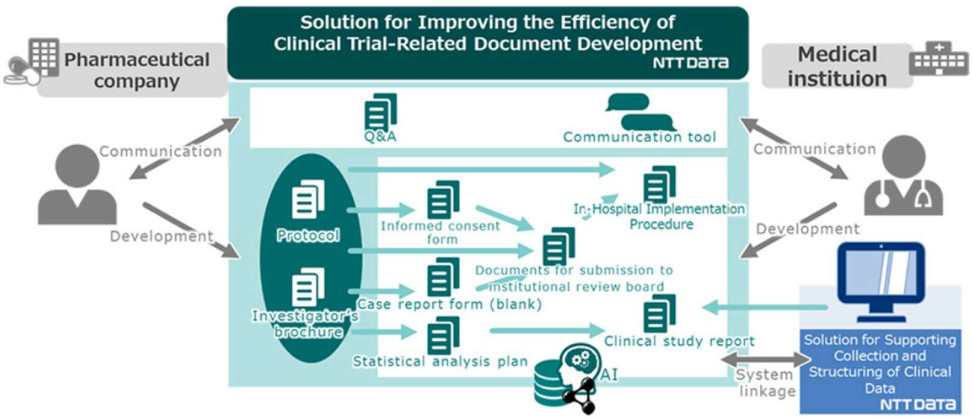

Figure 5: Overview of the solution for improving the efficiency of clinical trial-related document development

5. Dilemmas with Big Data Analysis in Chugai Pharmaceutical Co., Ltd.:

5.1. Social Dilemmas:

The risk of big data exacerbating inequality emphasizes Chugai's responsibility to promote equitable benefits across all patient populations. To achieve this, Chugai must analyze extensive patient and family data, data post-adjustment to recommended maintenance doses, and disease progression patterns. Every study has undergone analysis to tackle these issues (Shima et al., 2020). Chugai Group is committed to fostering a sustainable society by tackling social issues with inventive strategies and dedication to human rights, environmental protection, and other related causes. The group pursues cooperative resolutions and shared progress through transparent communication and positive relationships with all stakeholders (Chugai Pharmaceutical co., ltd., 2024).

5.2. Data Protection Laws:

On May 25, 2018, the General Data Protection Regulation (GDPR) replaces the Data Protection Act of 1998, impacting all organizations that gather, maintain, and process personal data of customers, employees, or individuals. Chugai is obligated to comply with data protection laws such as GDPR to safeguard patient data (Sample Data Protection Policy Template, 2018).

GDPR differentiates between personal data, known as special categories of data, and sensitive data, which poses risks to individuals and demands extra protection measures. Personal data includes any information gathered from individuals such as customers, employees, or contractors and must be managed according to GDPR guidelines. This involves implementing both organizational and technological safeguards for data protection. Regular reviews of GDPR compliance documents are crucial to ensure that privacy policies adhere to regulations, including clear retention policies for general and specific personal data categories. Furthermore, agreements with external parties handling personal data on behalf of the organization should be established, with all associated documentation updated and compliant with GDPR requirements (Tarasiuk, 2022).

5.3. Privacy and Confidentiality:

When big data is applied in healthcare, the importance of patient privacy protection increases significantly. Chugai's obligation entails respecting patient privacy, valuing individual autonomy, and securing permission to utilize patient data. The incorporation of big data presents ethical and legal challenges, posing considerable risks and issues, particularly in relation to patient privacy. Reducing these privacy risks entails expenses that influence not only existing patients but also the broader healthcare system (Price & Cohen, 2019).

Chugai Pharma strictly adheres to using collected personal data solely for necessary and legitimate purposes. This involves activities such as account creation, information dissemination, and account security maintenance. The sharing, transferring, or disclosing of personal information to third parties is strictly prohibited unless permitted by law, required by governmental authorities, or necessary to enhance services through affiliated entities during mergers and acquisitions, or for other legitimate purposes (Chugai Pharmaceutical co., ltd., 2023).

Conclusion:

In the conclusion, this study offers a detailed exploration of big data analytics integration within the Pharmaceutical Industry, addressing aspects like methodologies, advantages, hurdles, and ethical implications. The research's analysis of various analytics approaches and their effect on decision-making efficiency underscores big data's potential to revolutionize efficiency and innovation in the industry. The discussion of key benefits and possible solutions to challenges provides crucial insights for stakeholders aiming to effectively harness big data. Additionally,

the examination of ethical, legal, and social ramifications emphasizes the importance of responsible data management in pharmaceutical research. Overall, the study showcases big data analytics' potential to drive progress in pharmaceuticals while advocating for ethical and well-informed decision-making to maximize benefits and minimize risks for patients and society.

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